

CIRCLE THEOREMS

Higher | Mixed applications

Circle geometry with trigonometry, algebra and coordinates

Name

Date

12 questions

51 marks

2017-2025

Instructions

- Answer each question in the space provided.
- Show all your working and follow the rounding instructions.
- Diagrams are not accurately drawn unless stated otherwise.
- Use a calculator except for pages marked NON-CALCULATOR.

Using this booklet

Questions are arranged by exam series. Each page gives its original paper reference and question number. The question numbers within the extracted artwork are the original exam numbers.

Includes multi-step applications combining circle geometry with other GCSE skills.

This is an independently compiled practice resource, not an official Pearson examination paper. The source collection covers 2017-2025. Questions are included only in the booklet that matches their main purpose.

Question index

Higher | Mixed applications | 12 questions | 51 marks

Practice	Exam series	Paper	Original question	Marks	Page
01	June 2017	1MA1/2H	Q23	3	3
02	June 2017	1MA1/3H	Q18	5	4
03	November 2017	1MA1/2H	Q20	5	5
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12	November 2025	1MA1/2H	Q18	5	14

The original exam numbering is retained in each question.

Paper 1 questions are marked NON-CALCULATOR on their pages.

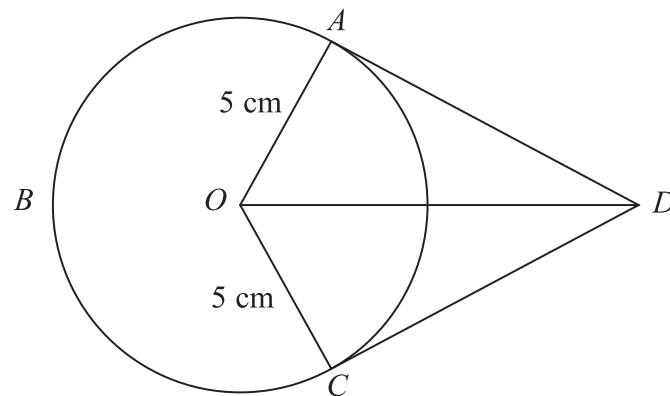
23 **L** is the circle with equation $x^2 + y^2 = 4$

$P\left(\frac{3}{2}, \frac{\sqrt{7}}{2}\right)$ is a point on **L**.

Find an equation of the tangent to **L** at the point P .

(Total for Question 23 is 3 marks)

18



A , B and C are points on a circle of radius 5 cm, centre O .

DA and DC are tangents to the circle.

$DO = 9$ cm

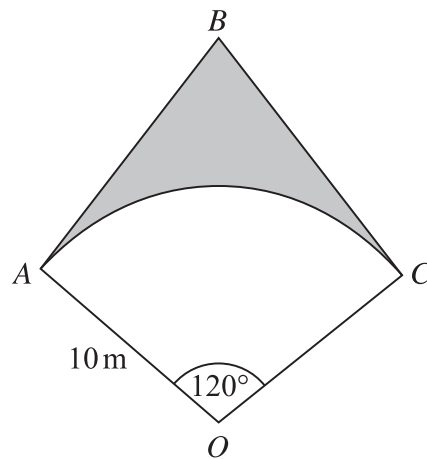
Work out the length of arc ABC .

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 18 is 5 marks)

20



OAC is a sector of a circle, centre O , radius 10 m .

BA is the tangent to the circle at point A .

BC is the tangent to the circle at point C .

Angle $AOC = 120^\circ$

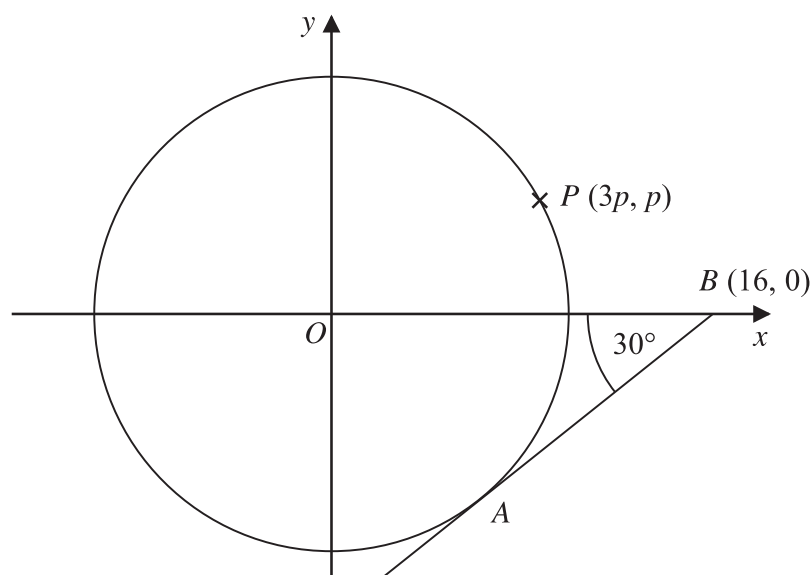
Calculate the area of the shaded region.

Give your answer correct to 3 significant figures.

..... m^2

(Total for Question 20 is 5 marks)

22 The diagram shows a circle, centre O .



AB is the tangent to the circle at the point A .
Angle $OBA = 30^\circ$

Point B has coordinates $(16, 0)$

Point P has coordinates $(3p, p)$

Find the value of p .

Give your answer correct to 1 decimal place.

You must show all your working.

$p = \dots\dots\dots$

(Total for Question 22 is 4 marks)

22 C is a circle with centre the origin.

A tangent to C passes through the points $(-20, 0)$ and $(0, 10)$

Work out an equation of C .

You must show all your working.

(Total for Question 22 is 5 marks)

20 The centre of a circle is the point with coordinates $(-1, 3)$

The point A with coordinates $(6, 8)$ lies on the circle.

Find an equation of the tangent to the circle at A .

Give your answer in the form $ax + by + c = 0$ where a , b and c are integers.

(Total for Question 20 is 4 marks)

24 A circle has equation $x^2 + y^2 = 12.25$

The point P lies on the circle.

The coordinates of P are $(2.1, 2.8)$

The line L is the tangent to the circle at point P .

Find an equation of L .

Give your answer in the form $ax + by = c$, where a , b and c are integers.

(Total for Question 24 is 4 marks)

23 A circle has equation $x^2 + y^2 = 25$

The point P with coordinates $(-3, 4)$ lies on the circle.

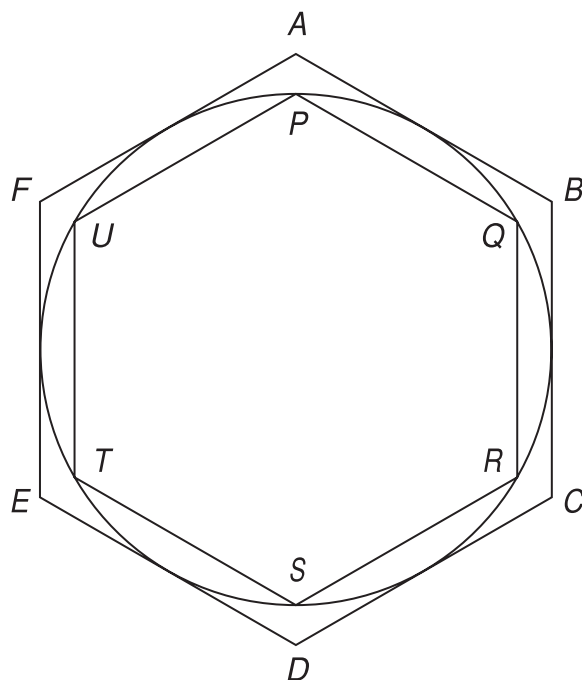
Alex says that the tangent to the circle at P crosses the x -axis at the point $(-8, 0)$

Is Alex correct?

You must show how you get your answer.

(Total for Question 23 is 4 marks)

21 The diagram shows a circle, radius r cm and two regular hexagons.



Each side of the larger hexagon $ABCDEF$ is a tangent to the circle.
Each side of the smaller hexagon $PQRSTU$ is a chord of the circle.

By considering perimeters, show that

$$3 < \pi < 2\sqrt{3}$$

(Total for Question 21 is 4 marks)

20 **C** is the circle with equation $x^2 + y^2 = 4$

Find an equation of the tangent to **C** at the point $(p, 1)$ where $p > 0$

Give your answer in the form $y + \sqrt{a}x = b$ where a and b are integers.

You must show all your working.

(Total for Question 20 is 4 marks)

Additional working space

22 C is a circle with centre $(0, 0)$

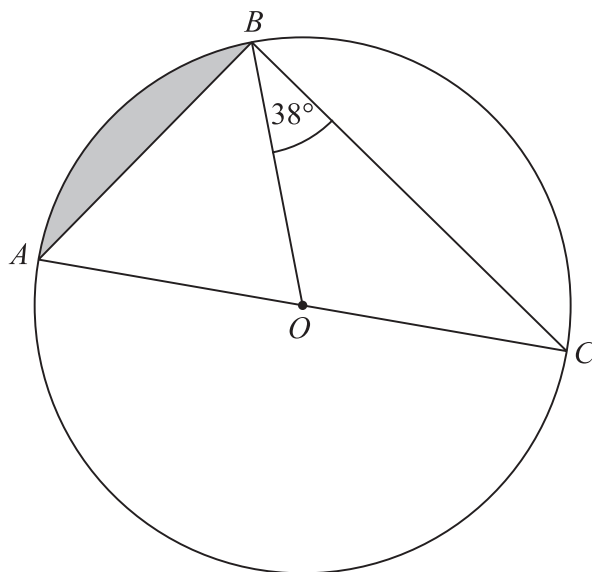
The straight line with equation $3x - 2y = 52$ is the tangent to C at the point P .

Find the coordinates of P .

(..... ,)

(Total for Question 22 is 4 marks)

18 A , B and C are points on a circle with centre O and radius 12 cm.



AOC is a diameter of the circle.

Angle $OBC = 38^\circ$

Calculate the area of the shaded segment.

Give your answer correct to 2 decimal places.

..... cm^2

(Total for Question 18 is 5 marks)